Pavan Kalyan Nemalapuri

B.Tech – (2023–27)
Electronics and Communication Engineering
KL University
2300040115ece@gmail.com | +91-8897758735 | linkedin.com/in/pavan-kalyan-nemalapuri

EDUCATION

KL University - B.Tech in Electronics and Communication Engineering, **9.76 CGPA** — July 2023 – May 2027
SR Junior College -Andhra Pradesh Board Of Intermediate, **98.3%** — June 2021 – March 2023
Sai Vineeth School – Board of Secondary Education Andhar Pradesh, **97.5%** — March 2021

Technical Skills

- **Programming:** C Programming, Java.
- **Hardware Description Language:** Verilog HDL.
- **EDA Tools:** Xilinx Vivado, LTspice ,Cadence.
- **Hardware Platforms:** Arduino Uno, ESP32, ESP8266, FPGA.

Internship

- **VLSI Internship Trainee – Skill Dzire (Virtual Mode)** — Jun 2025

  Worked on the design and implementation of a Priority Traffic Light Controller using Verilog HDL. Designed the system using FSM methodology with emphasis on synchronous design, timing behavior, and structured state transitions. Performed RTL simulation and functional verification as part of the digital design flow.

Projects

- **Automated Greenhouse Management System** — Sep 2025

  Designed an IoT-based greenhouse automation system to monitor temperature, humidity, and soil moisture in real time using ESP32 and environmental sensors. Implemented automated irrigation and climate control mechanisms based on threshold sensor values to improve water efficiency and crop monitoring. Integrated remote monitoring capability for continuous environmental data observation and system control.

- **Smart Parking Slot Detection System Using IR Sensors and Arduino** — Nov 2025

  Developed a smart parking management system using Arduino Uno and IR sensors to detect vehicle occupancy across multiple parking slots. Implemented embedded C logic to process digital sensor inputs and control LED indicators for real-time parking slot availability display. Designed the system to reduce manual parking management and improve parking space utilization efficiency.

- **Pipelined 16-bit Arithmetic Logic Unit (ALU) in Verilog** — Dec 2025

  Designed and implemented a 16-bit pipelined Arithmetic Logic Unit (ALU) using Verilog HDL to perform arithmetic and logical operations including addition, subtraction, AND, OR, XOR, and shift operations. Implemented a 2-stage pipeline architecture with synchronous registers to improve throughput and optimize timing performance. Developed modular ALU components and verified functionality through RTL simulation, testbench validation, and waveform analysis using simulation tools.

Certifications

- VLSI Design and Verification – TARAS Systems & Solutions
- Linguaskill Certificate – Cambridge University

Achievements

- **Merit Scholarship – KL University** Awarded 4.8 Lakhs merit scholarship for academic excellence in Intermediate, with 50 percentage tuition fee concession provided each semester.
- **Hackathon Participation – KL University** — 2025

  Participated in a university-level hackathon and presented a team-based technical solution focused on real-world problem solving. Improved practical knowledge in teamwork, idea development, and project presentation under competitive conditions.